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%Assessment 2                                :Geared-shaft Torsional Vibration System
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%Module                                      :SME3701

%%Given Parameters
d  = 0.050;                                  %Diameter in meters
L  = 1.00;                                   %Length in meters
G  = 79.3e9;                                 %Shear modulus in Pascals
J0 = 0.10;                                   %Mass moment of Inertia in kg.m^2
T  = 1000;                                   %Transmitted Torque in N.m
N_driven = 1000;                             %Driving speed in rpm
Teeth  = 16;                                 %Number of driving gear teeth

%System Parameter Calculation for Simulink

%Calculate Polar Moment of Inertia (J_p):
J_p = (pi / 32) * d^4;  % Polar moment of inertia = (pi / 32) * 0.05^4 =
6.136x10^-7 m^4

%Calculate Torsional Stiffness (k_t):
k_t = (G * J_p) / L;  % Torsional stiffness = (79.3e9 * 6.136x10^-7) / 1 =
48657.8 N.m/rad

%Calculate System's Natural Frequency (f_n):
omega_n = sqrt (k_t / J0); % omega_n = sqrt (48657.8 / 0.10) = 697.55 rad/s
% Calculate the system's natural frequency in Hz
f_n = omega_n / (2 * pi);  % Convert from rad/s to Hz = 697.55 / (2 * pi) =
111.02 Hz
%Nominal Driven Angular Velocity:
omega_driven = (2 * pi * N_driven) / 60; % omega_driven = (2 * pi * 1000) /
60 = 104.72 rad/s

%Calculate Static Deflection (theta) Under Steady Condition:
theta = T / k_t; % Static Deflection = 1000 / 48657.8 = 0.02055 rad

%Calculate Frequency of Driven Gear:
N_driven_Hz = N_driven / 60; % Frequency of driven gear = 1000 / 60 = 16.67
Hz
t_p = 1 / N_driven_Hz; % Pulse period per broken tooth = 1 / 16.67 = 0.06 s

%Calculate Tooth Mesh/Impact Frequency (f_"tooth"):
f_tooth = N_driven_Hz * Teeth; % Tooth mesh frequency = 16.67 * 16 = 266.67
Hz
T_impact = 1 / f_tooth; % Time between tooth impact = 1 / 266.67 = 0.00375 s



%1.2 Simulink scope video attached
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%1.3 a) Max angular displacement in rad = 27.23 rad
%1.3 b) Max angular velocity in rad = 0.0596 rad


%1.4 Natural frequency = 111.02 Hz (dominant torsional vibration)
%1.4 Tooth Mesh frequency = 266.67 Hz
%1.4 Driven gear frequency = 16.67 Hz
% Based on these different frequencies resonance does not occur, large
% spikes are not caused by resonance continuous build-up. Instead each time
% tooth hits every 0.06 sec it triggers a strong natural free vibration
% then gradually fades away before the next impact.
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https://github.com/Ziphozethu-Magqazolo/16622685/blob/main/16622685.pdf

https://drive.mathworks.com/files/b16622685.mlx

https://drive.mathworks.com/files/b16622685.slx

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